

Titre de la thèse

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List of Acronyms

FEM Finite Elements Method.

SCM Spectral Collocation Method.

SRR Split-Ring Resonator.

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Introduction

Context

Noise is a major issue in our modern societies, ranging from everyday life inconveniences, work comfort problems to an actual pollution problem in severely affected areas.

impact du bruit, enjeux psycho-santé / biodiversité / indus / travail ? rapport au bruit dans les cultures

voir la taille de la partie ?

State of art

1. State of art

- Acoustic isolation strategies
 - Insulation using foams
 - Rise of metamaterials: court historique
 - Classical metamaterials
 - multilayer, gradient index, controlled anisotropy etc.
 - metaporous
 - others
-
- porous / fibrous materials [1–6]. Largely used in the industry, they are efficient at frequencies starting close to the quarter wavelength resonance frequency of the layer and have been extensively analyzed in the literature
 - tuning of the microstructure, gradient properties, anisotropic materials: Recently, with the increase of new fabrication techniques based on 3D printing, the structure of the porous materials can be controlled and tuned to design materials with better absorption capabilities at low frequencies, based on gradient properties [7–9], or on anisotropic materials [10–13]
 - metamaterials in general, with periodic structures, ...

In order to increase the capabilities of acoustic absorption of porous materials, some studies involving porous layers embedding periodic arrays of inclusions were done [14–17]. These inclusions can have either resonances or not and they allow for an increased dissipation

Figure 1: Image d'un poreux ?

Figure 2: Figure exemple de métamatériau**Figure 3:** Figure de biblio métamatériaux, prendre qq configs

in the viscous regime. These structures are referred to as metaporous surfaces when used in reflection problems. The first works involving metaporous surfaces or interfaces were developed for transmission problems [18, 19]. In case of more than one inclusion per spatial period, several absorption peaks can be obtained for a particular arrangement along the porous thickness [15]. Nevertheless, this rapidly leads to a thick structure. Near the inclusion resonance frequency, chosen to be in the viscous regime, an absorption peak occurs [20]. The resonance frequency of the inclusions depends on the orientation of the resonators relative to the rigid boundary [16, 20]. Recently, 3D printed porous materials with embedded split-ring resonators have been modeled [12], designed and experimentally characterized showing perfect absorption peaks. Figure ??(a) shows an example of 3D printed sample with its corresponding absorption coefficient. Previous metaporous materials consider rigid frame porous as host medium. Extensions of these materials to consider poroelastic materials have been done (see Fig. ??(b)) [21]. Interesting properties were exhibited for thin elastic shell inclusions, and particularly the coupling between the trapped and the first volume mode leading to a high wide band absorption. It is worth noting here that different modelings based on Finite Element Methods (FEM) and Multiple Scattering Theory (MST) have been validated for the case of poroelastic waves saturated by air [22].

Acoustic metamaterials are structured materials made of resonant building blocks with resonance frequencies in the sub-wavelength regime (i.e., for frequencies which wavelength is much bigger than the typical size of the material) [23–25]. These resonances introduce strong dispersion and the properties of the system are dramatically affected. Together with the unavoidable losses, the resonant behaviour of metamaterials make them perfect candidates to design efficient absorbers at low frequencies with reduced dimensions [26, 27]. Balancing the dissipation and leakage of each resonance yields the recently described critical coupling. In this case, the reflection drops to zero [28]. Perfect absorbers can thus be designed for reflection problems for both monochromatic waves [26, 29] and broad range of frequencies [29–32]. Recently, resonators filled with 3D printed porous materials have been developed to design perfect and broadband absorption (see Fig. ??(d)) [33].

One of the initial interests of the acoustic metamaterials community was the possibility of producing deeps in the transmission spectra at very low frequencies. This has been tremendously developed in the literature and we cannot cite the full list of works developing such approach [34–37]. Metamaterials are based on resonant building blocks and the transmission is zero in the lossless case at their resonance frequency. In reality, losses cannot be avoided and there is always a small amount of the energy that is transmitted. Membrane type acoustic metamaterials [38–40], or acoustic metamaterials made by Helmholtz resonators [41–43] have been used to develop systems with high values of transmission loss for audible acoustics. Attached resonators to walls have been also used to reduce the transmission loss of panels considering the vibroacoustic coupling [44, 45].

For the case of mirror-symmetric structures, it is worth noting here that if only a monopolar resonator (Helmholtz) or a dipolar resonator (membrane) is used to absorb waves, only 50% of energy can be absorbed as shown in several articles [46, 47]. However, using a material made of slits loaded by an array of identical resonators creates an accumulation point close to the resonance frequency of the resonators allowing the accumulation of the resonances of the slit, thus placing together symmetric and antisymmetric resonances [38, 41, 48], although requiring for a compromise between absorption and thickness. A more complex approach

consists in designing degenerate resonators. By using combinations of membrane and cavity resonators [38, 49] and with slits loaded with Helmholtz resonators [47], quasi-total absorption can be reached. In these systems, each resonator manages half of the problem, absorbing each one half of the energy and allowing a perfect absorption peak: all the energy that impinges on the system is dissipated within the system, i.e., the energy is neither reflected nor transmitted. The other alternative to obtain perfect absorption consists in breaking the symmetry of the system. These systems can achieve unidirectional perfect absorption, i.e., perfect absorption can be obtained from one of the two incidence sides [46]. The principle of these systems consists in using two resonators, one that is used to reduce as much as possible the transmission and the other that impedance matches the system from one side [26, 29, 40]. This concept has been exploited to develop broadband perfect absorbers using a cascade of the previous process giving rise to a rainbow absorber [30] (see Fig.??(d)).

Motivations & Work Objectives

- modal behaviour, band structure
- impact of the surrounding radiating media

- démarches, etc.

Organisation and contents of the thesis

Introduction has set the context as well as an overview of the literature in which this work is situated. We saw that acoustic metamaterials have emerged in the last decades and propose noise mitigation structures that are much more efficient than the traditional foams used for sound insulation. Overall, a focus area has been identified towards identifying more precisely the modal behaviour of metaporous structures.

Chapter 1 targets the study of the propagation of guided waves in multilayer arbitrary structures with dissipative layers. A simple numerical method, the **Spectral Collocation Method (SCM)** is implemented and allows for the computation of the dispersion relation for these configurations.

Chapter 2 is focused on several numerical strategies for the modelling of scattering properties of metaporous/metaporoelastic structures. A global method is coupled to a **Finite Elements Method (FEM)**.

In Chapter 3, the modal behaviour of metaporous structure is studied. The fem method from the previous chapter is adapted to describe this problem. Results are presented for a rigid inclusion, as well as with embedded **Split-Ring Resonator (SRR)**

Figure 4: Diagram presenting the various tackled configurations in the chapters

Chapter 1

Theoretical aspects

This introductory chapter lays the foundation for the thesis, giving some notions of the theoretical aspects in order to guide the reader through the various chapters that will follow. At first, a state of the art of the literature is proposed, covering the topics of guided waves in dissipative layers, leaky modes in structures, metaporous configurations, ...

The present chapter is intended as an introduction to the fundamental concepts deemed necessary for the good understanding of the manuscript. Some theoretical aspects will be presented as well as some bibliography elements, which are useful for the context the present research takes place in.

1.1 Guided waves formalism for dispersion computation vs. scattering problem ?

1.1.1 Reflection and transmission coefficient, scattering from a structure

1.1.2 Guided waves: formalism in layers/guides

1.1.3 Energy radiation to a surrounding medium: leaky waves

petit état de l'art: électromag, et situation en acoustique

illustration dans un cas simple: visco lamb ?

1.2 Analytical models for the description of porous materials

historique ? DBM -> JCA -> Biot dissipation mechanisms in porous layers focus on the attenuation of the waves

1.2.1 Rigid-frame: DBM to JCA

Elastic motion of the skeleton: Biot theory with various formulation

1.3 Numerics

1.3.1 Plane wave modelling

1.3.2 TMM and variations

1.3.3 Numerics for homogeneous layers

1.3.4 FEM for more complex geometries

Brouillon à partir de là

thèses: natacha aberkane-gauthier, théo cavalieri, macaglione, mathieu gaborit,

Context and Motivation This thesis is dedicated to the numerical modeling of wave propagation in complex metaporous and poroelastic multilayered structures. These materials combine the dissipative characteristics of porous media with geometric or structural features typical of acoustic metamaterials, offering advanced control over sound transmission, reflection, and absorption.

Understanding and predicting the acoustic behavior of such systems is essential for designing efficient sound absorbers, acoustic coatings, and noise-reducing structures. In particular, this work focuses on the role of guided and leaky wave modes in shaping scattering properties, and on the development of efficient numerical strategie

- example of a porous layer with a rigid-backing, computed using pw.
- attenuation of waves in layers generally ?
- schéma guided modes, leaky modes
- poroelasticity models
- numerical methods
- various structures

Conclusion & Perspectives

1. conclusion générale de la thèse

- ?

2. perspectives:

- study PEM host configurations
- homogeneous PEM layer disp validation SCM & GM
- excitation of the modes: COMSOL & line source excitations
- experimental validation: porous with rigid inclusions, SLaTCoW or other method

Appendices

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